Rule-based Monetary Policy in China

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Motivation and Research Question

- **Motivation**
  - China: The most significant economy but its monetary policy remains under-studied.
  - Currently facing a severe post-COVID demand slowdown.
    - Housing market slump; Local government debt crisis.
    - CPI stagnating around 0%; PPI negative since late 2022.
  - The People's Bank of China (PBoC)'s policy response appears passive or constrained.
    - No explicit monetary policy rule to anchor expectations.
    - Gradual and limited cuts to interest rates and reserve requirements ratio.
- **Research Question:**
  - **How would the Chinese Economy have differed if the PBoC had followed an explicit, rule-based (e.g., NGDP and Inflation Targeting) monetary policy?**

Empirical Strategy

- **Methodology:** Use the **semi-structural method** (McKay and Wolf 2023; Caravello, McKay, and Wolf 2024) to simulate counterfactual monetary policy rules in China.
- This approach requires two "sufficient statistics" instead of a full structural model, making it robust to the Lucas Critique.
- **Statistic 1: Reduced-Form Projections**
  - **Purpose:** The economy's baseline forecast, capturing all "fundamental" shocks (fiscal, demand, etc.) without needing to identify them individually.
  - **Estimation:** We will estimate this using a **Reduced-Form VAR** on key Chinese macro variables.
  - **Time Period**: Quarterly data from 2015 to 2025; for robustness, we will also use 2020-2025.

Empirical Strategy

- **Statistic 2: Policy Causal Effects**
  - **Estimation Method 1:** High-Frequency Identification (HFI) approach, following Das and Song (2023).
    - **Identification:** They defined a "shock" as the change in 1-year Interest Rate Swaps (IRS) on the exact day of a PBoC policy announcement.
    - **Policy Tracked:** Price-based (MLF, LPR) and quantity-based (RRR) tools, Quarterly Report from PBoC, exchange rate regime reforms, etc.
  - **Alternative HFI:** He et al. (2023) applied the two-stage HFI method developed by Bu, Rogers, and Wu (2021) to China using the rates of various types of Nonnegotiable Certificates of Deposit (NCDs).
  - **Estimation Method 2:** C. D. Romer and D. H. Romer (2004) shocks constructed from estimated monetary policy rules by Chen, Xiao, and Zha (2025).
    - Price-based rule that changes interest rates identified through 7-day repo rate (R007).

Sample Results from the Great Recession (Caravello, McKay, and Wolf 2024)

The simulation solves for the optimal sequence of policy interventions (Statistic 2) required to steer the economy's baseline forecast (Statistic 1) to the new counterfactual target (Inflation, NGDP, etc.)

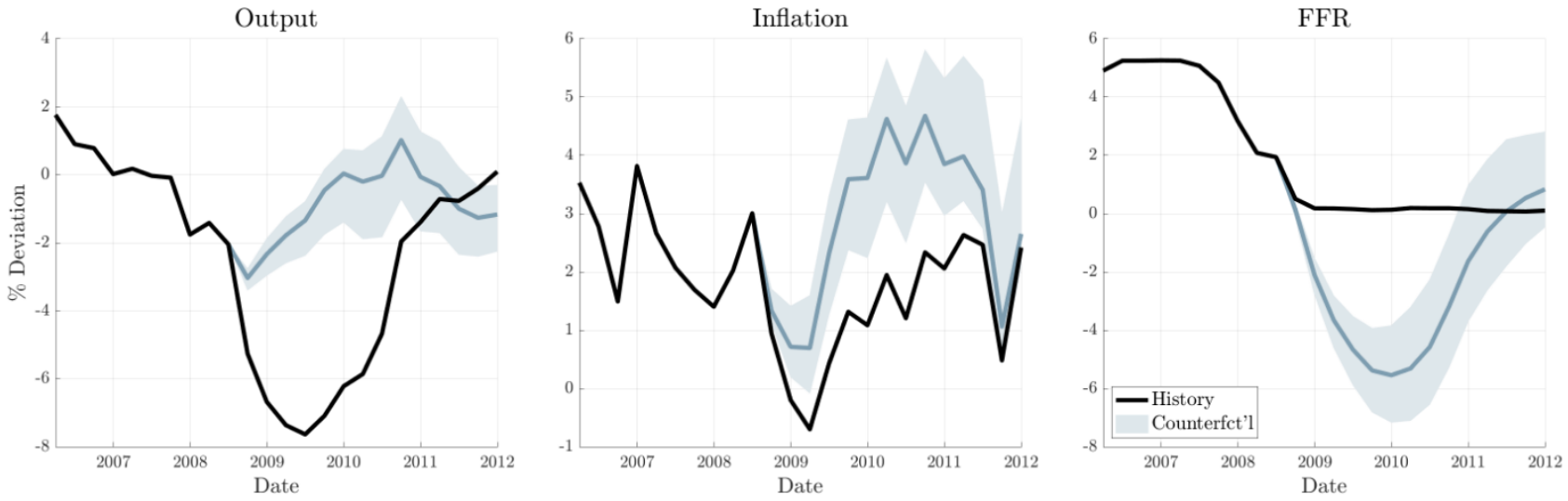

Figure 9: Counterfactual evolution of output, inflation, and the federal funds rate in the Great Recession, under the policy rule that minimizes (9) without any effective lower bound on rates. Black: data. Blue: posterior median (solid) and 16th and 84th percentile bands (shaded), drawing from posterior across all models and parameters.

Limitations: Structural Applicability to China

While the focus on the MLF rate addresses the "single instrument" assumption, three structural challenges remain:

- **1. The Fiscal-Monetary Nexus ("Separation Principle")**
  - *Model Assumption:* The private sector's structural equations are invariant to the policy rule.
  - *China Context:* Soft budget constraints (SOEs, LGFVs) create a quasi-fiscal channel. Aggressive counterfactual tightening could force structural breaks (e.g., fiscal bailouts), violating invariance.
- **2. Information Asymmetry ("Signaling Channel")**
  - *Model Assumption:* Policy actions do not reveal private information about the state of the economy.
  - *China Context:* Given lower data transparency, a PBOC rate hike might be interpreted as a signal of strong growth rather than tightening, reversing the intended effect.
- **3. Availability of "News Shocks"**
  - *Model Assumption:* Accurate counterfactuals require identifying "news shocks" (forward guidance) to span expectations.
  - *China Context:* Explicit forward guidance is historically limited. Weak identification of news shocks reduces the precision of ex-ante

Appendix

The Semi-Structural Method for Policy Counterfactuals

- **The Goal:** Evaluate policy counterfactuals (e.g., "what if the Fed had a different rule?") without committing to a fully specified structural model.
- **Core Insight:** A change in the policy *rule* is equivalent to an appropriate, expected path of policy *shocks* applied to the *baseline* economy forecast.
- **Surviving the Lucas Critique:**
  - The method models a systematic, fully expected rule change, not a sequence of surprises.
  - It uses a combination of date-0 shocks (contemporaneous + "news" shocks) to enforce the new rule in agents' expectations from the start.
- **Key "Structural" Assumption (The Split):**
  - The economy can be split into a **"non-policy block"** (private sector behavior) and a **"policy block"** (the rule).
  - The non-policy block is assumed to be stable and depends only on the *path of the policy instrument* (e.g., the yield curve), not on the rule that generates it.

Counterfactual Rules

1. **Strict Inflation Targeting:**

$$\min \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(\pi_t - \pi_t^*)^2 \right]$$

2. **Strict NGDP Targeting:**

$$\min \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(\Delta x_t - \Delta x_t^*)^2 \right]$$

3. **Flexible Inflation Targeting:**

$$\min \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(\pi_t - \pi_t^*)^2 + \lambda_y (y_t - y_t^*)^2 \right]$$

4. **Triple Mandate (PBoC-style):**
   - Includes the PBoC's explicit goal of currency stability.

$$\min \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[(\pi_t - \pi_t^*)^2 + \lambda_y (y_t - y_t^*)^2 + \lambda_e (e_t - e_t^*)^2 \right]$$

The Structural Part

- **Robustness Check / Future Work (The "Hybrid" Method):**
  - **Problem:** If our baseline fails (or as a check), we can address the critique that the empirical "Recipe" is incomplete (i.e., missing "long-end" effects).
  - **Check:** We can use a minimal structural model (e.g., a simple NK model) and "discipline" it by forcing its short-term IRF to match our HFI-estimated IRF but do not model shocks and monetary policy rules.
  - We then use this "disciplined" model to extrapolate the long-term part of our "Recipe" and re-run the simulation.

How to Measure Chinese Output

- **The Challenge:** Official quarterly GDP data is famously smoothed, low-frequency and over-revised.
- **Primary Approach:** We may use the The China Cyclical Activity Tracker developed by the San Francisco Fed (Fernald, Hsu, and Spiegel 2021).
- **Robustness Checks:**
  - Use other, well-measured monthly variables (like Industrial Production) as a simpler proxy.

References I

- Bu, Chunya, John Rogers, and Wenbin Wu (2021). "A Unified Measure of Fed Monetary Policy Shocks". In: *Journal of Monetary Economics* 118, pp. 331–349.
- Caravello, Tomás E., Alisdair McKay, and Christian K. Wolf (2024). *Evaluating Monetary Policy Counterfactuals: (When) Do We Need Structural Models?* Working Paper 32988. National Bureau of Economic Research.
- Chen, Kaiji, Yiqing Xiao, and Tao Zha (July 2025). *A Trade-off Between Monetary Policy Transmission and Systemic Risk in China*. Working Paper 34056. National Bureau of Economic Research.
- Das, Sonali and Wenting Song (2023). "Monetary Policy Transmission and Policy Coordination in China". In: *China Economic Review* 82, p. 102032.

References II

- Fernald, John G., Eric Hsu, and Mark M. Spiegel (2021). "Is China Fudging Its GDP Figures? Evidence from Trading Partner Data". In: *Journal of International Money and Finance* 110, p. 102262.
- He, Jianyao et al. (2023). "A High-Frequency Measure of Chinese Monetary Policy Shocks". In: *SSRN Electronic Journal*. Ahead of print.
- McKay, Alisdair and Christian K. Wolf (2023). "What Can Time-Series Regressions Tell Us About Policy Counterfactuals?" In: *Econometrica* 91.5, pp. 1695–1725.
- Romer, Christina D. and David H. Romer (2004). "A New Measure of Monetary Shocks: Derivation and Implications". In: *American Economic Review* 94.4, pp. 1055–1084.